



IFS Tobin Well Product

Data Dictionary – Tobin Data Portal

July 2024

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1. Introduction and Product Overview

This Data Dictionary provides IFS Tobin users a quick and easy-to-use tool to look up and associate field definitions. It allows for search by field or look up via each data layer format. Links outlined in [purple](#) provide users with the ability to jump to a field definition quickly within each data format. Links are also provided to code value tables and additional reference information.

The IFS Tobin Well Product currently covers 32 U.S. states and offshore Gulf of Mexico. It contains a comprehensive set of oil & gas wells and other associated wells which assist in the recovery of subsurface hydrocarbons. Data provided includes geographic locations (surface and bottom hole), notional well paths, and a rich complement of planning, operational, and dispositional attribution that coincides with the well's regulatory lifecycle.

For file geodatabase, shapefile, and CSV deliverables, the IFS Tobin Well Product is composed of five (5) data layers and two (2) related data tables. The spatial data layers include Well Surface, Well Surface Deleted, Well Bottom, Well Bottom Deleted, and Well Path. The related data tables include Well Surface Location and Well Bottom Location.

Beginning July 1, 2022, only this redesigned well schema is available in the Tobin Data Portal web application or through Tobin REST API. Beginning in November 2020, information displayed on the Tobin Data Portal web application or through the queries or identifies completed on the Tobin Data Portal is based on this redesigned well schema.

This product represents IFS Tobin's best effort in building this file based on information collected by IFS Tobin from various sources and contained internally at IFS Tobin. IFS Tobin makes no representations or warrants as to the completeness or accuracy of this product.

2. Field Definitions

Note: Code Field Names are displayed as [links](#) to Code Value Tables and Additional Reference Information

Field Name	Definition
Active Status Flag	<i>A flag, (Y/N), that indicates if the wellbore is currently active.</i>
Aliased Current Operator	<i>Alternate representation of Current Operator Name.</i>
Aliased Original Operator	<i>Alternate representation of Original Operator Name.</i>
API County Code	<i>A three-digit numeric code representing the county or pseudo-county identified in the well's API as recognized by the American Petroleum Institute's D12A API Well Number Standard and the PPDM US Well Number Standard.</i>
API Number (10)	<i>A ten-digit numeric value that uniquely identifies the well within a particular US state and county or pseudo-county as recognized by the American Petroleum Institute's D12A API Well Number Standard and the PPDM US Well Number Standard.</i>
API Number (12)	<i>A twelve-digit numeric value that uniquely identifies the wellbore within a particular US state and county or pseudo-county as recognized by the American Petroleum Institute's D12A API Well Number Standard and the PPDM US Well Number Standard.</i>
API State Code	<i>A two-digit numeric code representing the state or pseudo-state identified in the well's API as recognized by the American Petroleum Institute's D12A API Well Number Standard and the PPDM US Well Number Standard.</i>
BOREID	<i>IFS Tobin assigned unique identifier assigned to every wellbore.</i>
Bottom Hole X	<i>The X Cartesian coordinate value of the wellbore's bottom hole location. This value is dependent on the projection selected.</i>
Bottom Hole Y	<i>The Y Cartesian coordinate value of the wellbore's bottom hole location. This value is dependent on the projection selected.</i>
Coordinate System Code	<i>The European Petroleum Survey Group (EPSG) coordinate system code corresponding to the State Plane X and State Plane Y coordinates.</i>
County Name	<i>Name of the county in which the wellbore's surface or bottom hole is geographically located.</i>

Field Name	Definition
Created Date	<i>The date the record was first established in IFS Tobin's well data system.</i>
Current Completion Date	<i>The first date after a well was perforated and capable of producing from the most recent formation interval. The date may or may not be the same as perforation date.</i>
Current Operator	<i>The company name that owned and operated the well at the time it was last evaluated by IFS Tobin.</i>
Feature ID	<i>ESRI Object ID.</i>
Field ID	<i>The US Energy Information Administration (EIA) Field Code of the oil or gas field that the wellbore is geographically tied to.</i>
Field Name	<i>The US Energy Information Administration (EIA) Field Name of the oil or gas field that the wellbore is geographically tied to.</i>
Final Status Code	<i>This code is derived from a composition of Well Status, Direction, Product, Class and Completion Count</i>
Final Status Description	<i>Description of Final Status Code, e.g. Abandoned Well.</i>
First Completion Date	<i>The initial date after a wellbore was perforated and capable of producing.</i>
First Direction	<i>The cardinal direction of the First Distance.</i>
First Distance	<i>Distance, typically in feet, from a known reference line or point used to determine location of the wellbore's surface or bottom hole when described using line calls, perpendiculars, or footages. This represents the back-calculated line calls to the Tobin Grid when a known-point was used to spot the wellbore's surface or bottom hole location.</i>
First Elevation	<i>A wellbore reference point elevation from mean sea level (MSL) of the type described by First Elevation Type Code, given in feet.</i>
First Elevation Type Code	<i>The starting reference point on a drill site used for surface and downhole measurements associated with the First Elevation, (e.g. KB, GR).</i>
First Elevation Type Description	<i>A textual description of the starting reference type (e.g. Kelly Bushing, Ground Elevation) associated with the First Elevation Code.</i>
First Production Date	<i>The date the wellbore initially started producing.</i>
Formation Name	<i>The name assigned to the sedimentary rock formation the current producing completion in a wellbore is associated with.</i>
Ground Elevation	<i>The topographic elevation, in feet, at the proposed or actual surface hole location.</i>

Field Name	Definition
Ground Elevation Source	<i>A code that indicates the government entity, organization, document, or other resource where the ground elevation was obtained for the surface location.</i>
Ground Elevation Type Code	<i>The topographic reference type (e.g. GR, ES) associated with the Ground Elevation.</i>
Ground Elevation Type Description	<i>The textual description of the topographic reference type (e.g. Ground Elevation, Estimated) associated with the Ground Elevation Code.</i>
H2S Well	<i>A flag, (Y/N), that indicates if hydrogen sulfide is present.</i>
KB Elevation	<i>The distance from the Kelly Bushing to mean sea level (MSL) found by adding the ground elevation to the Kelly Bushing height.</i>
KB Elevation Source	<i>A code that indicates the government entity, organization, document, or other resource where the KB elevation was obtained.</i>
Last Production Date	<i>The date the wellbore last produced.</i>
Latitude	<i>The North American Datum 1927 (NAD27) latitude associated with the wellbore's surface or bottom hole location.</i>
Lease Name	<i>The operator assigned name that typically identifies the parent oil or gas lease(s) that a wellbore is permitted upon. In some states, this may instead be reported as Well Name, Farm Name, or Unit Name.</i>
Longitude	<i>The North American Datum 1927 (NAD27) longitude associated with the wellbore's surface or bottom hole location.</i>
Meridian	<i>The Principal Meridian that acts as the center line for the survey (grid) in which the wellbore's surface or bottom hole is geographically located.</i>
Minor Civil Division	<i>The primary governmental/administrative division as summarized by the federal government in which the wellbore's surface or bottom hole location resides.</i>
Object ID	<i>An Object ID is a unique, not null, integer field used to uniquely identify rows in tables in a geodatabase. Object IDs are limited to 32-bit values that store a maximum value of 2,147,483,647.</i>
Offshore Area	<i>The offshore protraction area created by the Bureau of Ocean Energy Management in which the wellbore's surface or bottom hole location resides.</i>
Offshore Block	<i>The offshore protraction area, created by the Bureau of Ocean Energy Management, further</i>

Field Name	Definition
	<i>subdivided into blocks three (3) miles by three (3) miles in size, in which the wellbore's surface or bottom hole location resides. The blocks are uniquely numbered within each protraction area.</i>
Original Operator	<i>The operator who permitted the well. If it is unknown, then the operator responsible for drilling and/or completing the well as supported by a directional survey, logs, IP test, core reports, rig reports, first production report, etc. If there is no supporting documentation, this field will be left empty (NULL).</i>
Permit Date	<i>The date the original permit to drill was submitted to or approved by the regulatory body depending on the state. This may also reflect the earliest available permit on record for the wellbore if the original is unavailable.</i>
Permit Expiration Date	<i>The calculated or explicit expiration date for the most recent approved drilling permit for the well prior to the wellbore's spud date.</i>
Plug Date	<i>The date a wellbore was Plugged and Abandoned.</i>
Plug Depth	<i>The plugging depth to permanently or temporarily close the well or wellbore as reported from the surface to the top of the plug.</i>
Proposed Flag	<i>The flag code used to indicate if the wellbore's surface or bottom hole coordinates and geographic location are "proposed" (P) or "actual" (A).</i>
Proposed MD	<i>The estimated total measured depth to which a wellbore will be drilled.</i>
Quad Name	<i>The USGS 7.5-minute topographic quadrangle map name where the wellbore's surface or bottom hole location is geographically positioned.</i>
Range	<i>The range distance from the Principal Meridian of the survey (grid) in which the wellbore's surface or bottom hole is geographically located.</i>
Range Direction	<i>The range direction (East or West) from the Principal Meridian of the grid in which the wellbore's surface or bottom hole is geographically located.</i>
Reference Detail	<i>The initial reference point, if any, utilized when determining a wellbore's surface or bottom hole location. For example, the corner of the grid or the name of a monument. This corresponds directly to First Distance, First Direction, Second Distance, Second Direction.</i>
Reference Type Code	<i>Code representing the type of reference line or point used as part of line calls, footages, bearing</i>

Field Name	Definition
	<i>and distance, or perpendiculars (e.g. A, SL). This corresponds directly to First Distance, First Direction, Second Distance, Second Direction.</i>
Reference Type Description	<i>The textual description of the type of reference line or point used as part of line calls, footages, bearing and distance, or perpendiculars (e.g. API, Surface).</i>
Regulatory Field ID	<i>The US Energy Information Administration (EIA) Regulatory Field Code of the oil or gas field that the wellbore is geographically tied to if the Regulatory Field Name is in the EIA master code field list.</i>
Regulatory Field Name	<i>The name assigned by the regulatory agency to an area either for the surface above hydrocarbons or a formation and area combination.</i>
Regulatory ID Number	<i>The permit number, well file number, serial number, or other identifier associated with the most recent successful completion or impending original completion of the wellbore. This identifier takes on many forms depending on the state or federal regulatory agency's internal requirements.</i>
Regulatory Lease Identifier	<i>The key value for a lease associated with the wellbore assigned by the regulatory agency.</i>
Related API	<i>This represents the corrected API replacing the previous API for the deleted surface or bottom. This is found only in the deleted surface or bottom hole layers.</i>
Second Direction	<i>The cardinal direction of the Second Distance.</i>
Second Distance	<i>Distance, typically in feet, from a known reference line or point used to determine location of the wellbore's surface or bottom hole when described using line calls, perpendiculars, or footages. This represents the back-calculated line calls to the Tobin Grid when a known-point was used to spot the wellbore's surface or bottom hole location.</i>
Second Elevation	<i>A wellbore reference point elevation from mean sea level (MSL) of the type described by Second Elevation Type Code, given in feet.</i>
Second Elevation Type Code	<i>The starting reference point on a drill site used for surface and downhole measurements associated with the Second Elevation, (e.g. KB, GR).</i>
Second Elevation Type Description	<i>A textual description of the starting reference type (e.g. Kelly Bushing, Ground Elevation) associated with the Second Elevation Code.</i>
Section	<i>The Public Land Survey System Section number assigned to the grid parcel in which the wellbore's surface or bottom hole location resides.</i>

Field Name	Definition
Section Quarters	<i>The Public Land Survey System or Carter section aliquot, or quarter, in which the wellbore's surface or bottom hole location resides.</i>
Shape	<i>Feature geometry (Polyline/Point).</i>
Shape Length	<i>Programmatically generated length of line.</i>
Sidetrack Number	<i>A two-digit numeric value that uniquely identifies each wellbore associated with a well as recognized by the American Petroleum Institute's D12A API Well Number Standard's API "Sidetrack Code" recognized by the PPDM US Well Number Standard's API "Wellbore Code".</i>
Spotted Location Description	<i>A textual description of a wellbore's surface or bottom hole location typically in the form of grid perpendiculars or geographic coordinates.</i>
Spotting Method Code	<i>Code referencing the type of legal used to spot the wellbore's surface or bottom hole location (e.g. AL, DD).</i>
Spotting Method Description	<i>A textual description of the type of legal used to spot the wellbore's surface or bottom hole location (e.g. Along a Line, Direction Distance).</i>
Spud Date	<i>The date the original drilling effort commenced.</i>
State District Name	<i>The Eastern State District name (colonial district) as defined by the state in which the wellbore's surface or bottom hole location resides.</i>
State Name	<i>Name of the state in which the wellbore's surface or bottom hole location resides.</i>
Surface X	<i>The X Cartesian coordinate value of the wellbore's surface location. This value is dependent on the projection selected.</i>
Surface Y	<i>The Y Cartesian coordinate value of the wellbore's surface location. This value is dependent on the projection selected.</i>
Survey Type Code	<i>Code that indicates the land grid type based on the wellbore's surface or bottom hole location in the IFS Tobin grid.</i>
Survey Type Description	<i>A textual description of the code that indicates the land grid type based on the wellbore's surface or bottom hole location in the IFS Tobin grid.</i>
Texas Abstract	<i>The Texas Abstract in which the wellbore's surface or bottom hole is geographically located.</i>
Texas Block	<i>The Texas land grid block in which the wellbore's surface or bottom hole is geographically located.</i>
Texas Grantee	<i>The entity in Texas to whom the designated survey was granted in which the wellbore's surface or bottom hole is geographically located.</i>

Field Name	Definition
Texas Labor	<i>The Texas labor value in which the wellbore's surface or bottom hole is geographically located.</i>
Texas League	<i>The Texas league value in which the wellbore's surface or bottom hole is geographically located.</i>
Texas Lot Number	<i>The Texas land grid/subdivision lot in which the wellbore's surface or bottom hole is geographically located.</i>
Texas Section	<i>The Texas section number in which the wellbore's surface or bottom hole location resides.</i>
Texas Survey Name	<i>The name of the Texas survey parcel in which the wellbore's surface or bottom hole is geographically located.</i>
Texas Tract Number	<i>The Texas tract number where the wellbore's surface or bottom hole location is geographically positioned.</i>
Total Depth	<i>The deepest measured depth to which the wellbore has been drilled.</i>
Township	<i>The Township distance from the baseline of the grid in which the wellbore's surface or bottom hole is geographically located.</i>
Township Direction	<i>The Township direction (North or South) from the baseline of the grid in which the wellbore's surface or bottom hole is geographically located.</i>
True Vertical Depth	<i>The straight-line, vertical distance from the wellbore's most recently completed bottom hole location and the reference point (GL, KB, or DF) as reported in the directional well survey or other associated well documents.</i>
Updated Date	<i>The last date the wellbore was updated or saved by IFS Tobin. This indicates that tabular, spatial, or both have been changed.</i>
WCS Update Date	<i>The date in which the wellbore record was last updated in the WCS delivery database.</i>
Well Bore Type Code	<i>Code that describes the drilling effort associated with the wellbore (e.g. OH – original hole, ST- side track).</i>
Well Bore Type Description	<i>The textual description of the code that describes the drilling effort associated with the wellbore (e.g. Original Hole, Sidetrack).</i>
Well Class Code	<i>Code that indicates general Lahee well classification of the wellbore (e.g. CT – core test).</i>
Well Class Description	<i>The textual description of the code that indicates general Lahee well classification of the wellbore (e.g. Core Test).</i>
Well Direction Code	<i>Code that indicates the movement direction of the wellbore's target fluid (product) at the time the well</i>

Field Name	Definition
	<i>was last evaluated by IFS Tobin (e.g. I – injection, H - huff and puff).</i>
<u>Well Direction Description</u>	<i>The textual description of the code that indicates the movement direction of the wellbore's target fluid (product) at the time the well was last evaluated by IFS Tobin (e.g. Injection, Huff and Puff).</i>
Well Number	<i>A well identifier typically assigned by the operator to the parent bore's surface hole in order to distinguish wells on a lease, farm, unit, or other grouping.</i>
<u>Well Product Code</u>	<i>Code that represents the target fluid of the wellbore, either produced or injected, at the time the well was last evaluated by IFS Tobin (e.g. G – gas, OG – oil-gas).</i>
<u>Well Product Description</u>	<i>The textual description of the code that represents the target fluid of the wellbore, either produced or injected, at the time the well was last evaluated by IFS Tobin (e.g. Gas, Oil-Gas).</i>
<u>Well Profile Code</u>	<i>Code that indicates the general trajectory of the wellbore relative to the vertical plane (e.g. V – vertical, D – directional).</i>
<u>Well Profile Description</u>	<i>The textual description of the code that indicates the general trajectory of the wellbore relative to the vertical plane (e.g. Vertical, Directional).</i>
<u>Well Status Code</u>	<i>The operational status of the associated wellbore as last evaluated by IFS Tobin (e.g. AB – abandoned, S – shut in).</i>
<u>Well Status Description</u>	<i>The textual description of the operational status of the associated wellbore as last evaluated by IFS Tobin (e.g. Abandoned, Shut in).</i>
WELLID	<i>IFS Tobin unique identifier assigned to every well at the surface level. Wellbores of the same parent well will share the same WELLID.</i>
<u>Zone Code</u>	<i>The State Plane Zone code associated with the wellbore's State Plane X and State Plane Y surface or bottom hole coordinates.</i>

3. Tobin Data Portal Product Formats – IFS Tobin Well Product

The IFS Tobin Well Product may be downloaded from the Tobin Data Portal in five (5) formats: file geodatabase, shapefile, comma-separated values, comma-separated values combined, or WCS.

For the file geodatabase deliverable, the IFS Tobin Well Product includes the following files:

File Geodatabase File Type	File Geodatabase File Name
<i>Spatial Layer</i>	<i>well_surface</i>
<i>Tabular Layer</i>	<i>well_surface_location</i>
<i>Relationship Class</i>	<i>well_surface_well_surface_location</i>
<i>Spatial Layer</i>	<i>well_surface_deleted</i>
<i>Spatial Layer</i>	<i>well_bottom</i>
<i>Tabular Layer</i>	<i>well_bottom_location</i>
<i>Relationship Class</i>	<i>well_bottom_well_bottom_location</i>
<i>Spatial Layer</i>	<i>well_bottom_deleted</i>
<i>Spatial Layer</i>	<i>well_path</i>

See [3.1 File Geodatabase \(FGDB\)](#).

For the shapefile deliverable, the IFS Tobin Well Product includes the following files:

Shapefile File Type	Shapefile File Name
<i>Spatial Layer</i>	<i>well_surface</i>
<i>Tabular Layer</i>	<i>well_surface_location</i>
<i>Spatial Layer</i>	<i>well_surface_deleted</i>
<i>Spatial Layer</i>	<i>well_bottom</i>
<i>Tabular Layer</i>	<i>well_bottom_location</i>
<i>Spatial Layer</i>	<i>well_bottom_deleted</i>
<i>Spatial Layer</i>	<i>well_path</i>

See [3.2 Shapefile \(SHP\)](#).

For the comma-separated values deliverable, the IFS Tobin Well Product includes the following files: well surface – well_surface, well surface location – well_surface_location, well surface deleted – well_surface_deleted, well bottom – well_bottom, well bottom location – well_bottom_location, and well bottom deleted – well_bottom_deleted. See [3.3 Comma-Separated Values \(CSV\)](#).

For the comma-separated values combined deliverable, the IFS Tobin Well Product includes the following file: well_combined. See [3.4 Comma-Separated Values Combined \(CSV\)](#).

3.1 File Geodatabase (FGDB)

The file geodatabase (FGDB) deliverable for the IFS Tobin Well Product is composed of a spatial layer file – *well_surface*, an associated tabular layer file – *well_surface_location*, and a relationship class – *well_surface_well_surface_location* for the well surface hole locations, a spatial layer file – *well_bottom*, an associated tabular layer file – *well_bottom_location*, and a relationship class – *well_bottom_well_bottom_location* for the well bottom hole locations, and a spatial layer file representing the path between the surface hole locations and bottom hole locations – *well_path*. There are also two (2) spatial layer files for deleted surface hole locations – *well_surface_deleted* and bottom hole locations – *well_bottom_deleted*.

3.1.1 File Geodatabase (FGDB) – Well Surface Location, Spatial

This is the spatial layer related to well surface hole locations for the IFS Tobin Well Product: *well_surface*. This file includes pertinent “point in time” well header information including API 10, operator, ground elevation, etc., allowing for analysis of the spatial, physical, and lifecycle characteristics of the well.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
WELL_NUM	Well Number	Text (20)
LEASE_NAME	Lease Name	Text (60)
CURRENT_OPERATOR	Current Operator	Text (60)
CURRENT_OPERATOR_ALIAS	Aliased Current Operator	Text (60)
ORIGINAL_OPERATOR	Original Operator	Text (60)
ORIGINAL_OPERATOR_ALIAS	Aliased Original Operator	Text (60)
STATE_CODE	API State Code	Numeric (Short)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Short)
COUNTY_NAME	County Name	Text (60)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
SURFACE_X	Surface X	Numeric (Double)
SURFACE_Y	Surface Y	Numeric (Double)
COORDINATE_SYSTEM_CODE	Coordinate System Code	Numeric (Long)
ZONE_CODE	Zone Code	Text (5)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
PERMIT_DATE	Permit Date	Date (no timestamp)

Exported Name	Field Name	Type
PERMIT_EXPIRY_DATE	Permit Expiration Date	Date (no timestamp)
SPUD_DATE	Spud Date	Date (no timestamp)
FIRST_COMPLETION_DATE	First Completion Date	Date (no timestamp)
ACTIVE_STATUS_FLAG	Active Status Flag	Text (1)
PROPOSED_FLAG	Proposed Flag	Text (1)
GROUND_ELEVATION	Ground Elevation	Numeric (Double)
GROUND_ELEVATION_TYPE_CODE	Ground Elevation Type Code	Text (10)
GROUND_ELEVATION_TYPE	Ground Elevation Type Description	Text (40)
H2S_FLAG	H2S Well	Text (1)
OBJECT_ID	Object ID	Object ID
SHAPE	Shape	Geometry

3.1.1.1 File Geodatabase (FGDB) – Well Surface Location, Tabular

This is the tabular layer related to well surface hole locations for the IFS Tobin Well Product: *well_surface_location*. This table includes survey and directional information and has a relational link to the Well Surface Location spatial layer. This table also has an associated relationship class: *well_surface_well_surface_location*.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
FIRST_DIRECTION	First Direction	Text (15)
FIRST_DISTANCE	First Distance	Numeric (Long)
SECOND_DIRECTION	Second Direction	Text (15)
SECOND_DISTANCE	Second Distance	Numeric (Long)
SPOTTED_METHOD_CODE	Spotting Method Code	Text (10)
SPOTTED_METHOD_DESCRIPTION	Spotting Method Description	Text (30)
SPOTTED_LOCATION	Spotted Location Description	Text (1024)
SURVEY_TYPE_CODE	Survey Type Code	Text (10)
SURVEY_TYPE	Survey Type Description	Text (30)
MINOR_CIVIL_DIVISION	Minor Civil Division	Text (60)
STATE_DISTRICT_NAME	State District Name	Text (60)
TX_ABSTRACT	Texas Abstract	Text (8)

Exported Name	Field Name	Type
TX_GRANTEE	Texas Grantee	Text (60)
TX_SURVEY_NAME	Texas Survey Name	Text (60)
TX_BLOCK	Texas Block	Text (30)
TX_SECTION	Texas Section	Text (40)
TX_LEAGUE	Texas League	Text (8)
TX_LABOR	Texas Labor	Text (8)
TX_TRACT_NUMBER	Texas Tract Number	Text (132)
TX_LOT_NUMBER	Texas Lot Number	Text (132)
MERIDIAN	Meridian	Numeric (Long)
TOWNSHIP	Township	Text (6)
TOWNSHIP_DIRECTION	Township Direction	Text (1)
RANGE	Range	Text (6)
RANGE_DIRECTION	Range Direction	Text (1)
SECTION	Section	Text (40)
SECTION_QUARTERS	Section Quarters	Text (132)
OFFSHORE_AREA	Offshore Area	Text (60)
OFFSHORE_BLOCK	Offshore Block	Text (60)
QUAD_NAME	Quad Name	Text (60)
REFERENCE_DETAIL	Reference Detail	Text (132)
REFERENCE_TYPE_CODE	Reference Type Code	Text (3)
REFERENCE_TYPE	Reference Type Description	Text (30)
Objectid	Object ID	Object ID

3.1.2 File Geodatabase (FGDB) – Well Surface Deleted, Spatial

This is the spatial layer related to well surface hole locations that have been deleted from the IFS Tobin Well Product: *well_surface_deleted*. The wells have been deleted because they are duplicates or temporary APIs or they are not part of the Oil & Gas process, i.e. cathodic wells, water wells – personal use, uranium wells, etc.

Exported Name	Field Name	Type
WellID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
STATE_CODE	API State Code	Numeric (Long)

Exported Name	Field Name	Type
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Long)
COUNTY_NAME	County Name	Text (60)
RELATED_API	Related API	Text (30)
LEASE_NAME	Lease Name	Text (60)
WELL_NUM	Well Number	Text (60)
FIELD_NAME	Field Name	Text (30)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
OBJECT_ID	Object ID	Object ID
SHAPE	Shape	Geometry

3.1.3 File Geodatabase (FGDB) – Well Bottom Location, Spatial

This is the spatial layer related to wellbore bottom hole locations for the IFS Tobin Well Product: *well_bottom*. This file includes pertinent “point in time” well information including API 12, operator, permit, spud, and completion dates, KB and reference elevations, producing product, well status, etc., allowing for analysis of the spatial, physical, and lifecycle characteristics of the wellbore.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
BOREID	BOREID	Numeric (Long)
API12	API Number (12)	Text (12)
API10	API Number (10)	Text (10)
SIDE_TRACK_NUMBER	Sidetrack Number	Text (2)
REGULATORY_LEASE_ID	Regulatory Lease Identifier	Text (60)
REGULATORY_ID	Regulatory ID Number	Text (60)
WELL_NUM	Well Number	Text (20)
LEASE_NAME	Lease Name	Text (60)
CURRENT_OPERATOR	Current Operator	Text (60)
CURRENT_OPERATOR_ALIAS	Aliased Current Operator	Text (60)
ORIGINAL_OPERATOR	Original Operator	Text (60)

Exported Name	Field Name	Type
ORIGINAL_OPERATOR_ALIAS	Aliased Original Operator	Text (60)
STATE_CODE	API State Code	Numeric (Long)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Long)
COUNTY_NAME	County Name	Text (60)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
BH_X	Bottom Hole X	Numeric (Double)
BH_Y	Bottom Hole Y	Numeric (Double)
COORDINATE_SYSTEM_CODE	Coordinate System Code	Numeric (Long)
ZONE_CODE	Zone Code	Text (5)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
PERMIT_DATE	Permit Date	Date (no timestamp)
PERMIT_EXPIRY_DATE	Permit Expiration Date	Date (no timestamp)
SPUD_DATE	Spud Date	Date (no timestamp)
FIRST_COMPLETION_DATE	First Completion Date	Date (no timestamp)
CURRENT_COMPLETION_DATE	Current Completion Date	Date (no timestamp)
FIRST_PRODUCTION_DATE	First Production Date	Date (no timestamp)
LAST_PRODUCTION_DATE	Last Production Date	Date (no timestamp)
PLUG_DATE	Plug Date	Date (no timestamp)
ACTIVE_STATUS_FLAG	Active Status Flag	Text (1)
PROPOSED_FLAG	Proposed Flag	Text (1)
FIRST_ELEVATION	First Elevation	Numeric (Double)
FIRST_ELEVATION_TYPE_CODE	First Elevation Type Code	Text (10)
FIRST_ELEVATION_TYPE	First Elevation Type Description	Text (15)
SECOND_ELEVATION	Second Elevation	Numeric (Double)
SECOND_ELEVATION_TYPE_CODE	Second Elevation Type Code	Text (10)
SECOND_ELEVATION_TYPE	Second Elevation Type Description	Text (15)
FIELD_ID	Field ID	Numeric (Long)
FIELD_NAME	Field Name	Text (100)

Exported Name	Field Name	Type
REGULATORY_FIELD_ID	Regulatory Field ID	Text (30)
REGULATORY_FIELD_NAME	Regulatory Field Name	Text (50)
FORMATION_NAME	Formation Name	Text (200)
PROPOSED_MD	Proposed MD	Numeric (Double)
KB_ELEVATION	KB Elevation	Numeric (Double)
TOTAL_DEPTH_MD	Total Depth	Numeric (Double)
TOTAL_DEPTH_TVD	True Vertical Depth	Numeric (Double)
PLUG_DEPTH	Plug Depth	Numeric (Double)
FINAL_STATUS_CODE	Final Status Code	Text (3)
FINAL_STATUS	Final Status Description	Text (30)
BORE_TYPE_CODE	Well Bore Type Code	Text (3)
BORE_TYPE	Well Bore Type Description	Text (30)
BORE_DIRECTION_CODE	Well Direction Code	Text (3)
BORE_DIRECTION	Well Direction Description	Text (30)
PROFILE_CODE	Well Profile Code	Text (3)
PROFILE	Well Profile Description	Text (30)
CLASS_CODE	Well Class Code	Text (3)
CLASS	Well Class Description	Text (30)
PRODUCT_CODE	Well Product Code	Text (3)
PRODUCT	Well Product Description	Text (30)
STATUS_CODE	Well Status Code	Text (3)
STATUS	Well Status Description	Text (30)
SHAPE	Shape	Geometry
OBJECT_ID	Object ID	Object ID

3.1.3.1 File Geodatabase (FGDB) – Well Bottom Location, Tabular

This is the tabular layer related to wellbore bottom hole locations for the IFS Tobin Well Product: *well_bottom_location*. This table includes survey and directional information and has a relational link to the Well Bottom Location spatial layer. This table also has an associated relationship class: *well_bottom_well_bottom_location*.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
BOREID	BOREID	Numeric (Long)
API10	API Number (10)	Text (10)
API12	API Number (12)	Text (12)
FIRST_DIRECTION	First Direction	Text (15)
FIRST_DISTANCE	First Distance	Numeric (Long)
SECOND_DIRECTION	Second Direction	Text (15)
SECOND_DISTANCE	Second Distance	Numeric (Long)
SPOTTED_METHOD_CODE	Spotting Method Code	Text (10)
SPOTTED_METHOD_DESCRIPTION	Spotting Method Description	Text (30)
SPOTTED_LOCATION	Spotted Location Description	Text (1024)
SURVEY_TYPE_CODE	Survey Type Code	Text (10)
SURVEY_TYPE	Survey Type Description	Text (30)
MINOR_CIVIL_DIVISION	Minor Civil Division	Text (60)
STATE_DISTRICT_NAME	State District Name	Text (60)
TX_ABSTRACT	Texas Abstract	Text (8)
TX_GRANTEE	Texas Grantee	Text (60)
TX_SURVEY_NAME	Texas Survey Name	Text (60)
TX_BLOCK	Texas Block	Text (30)
TX_SECTION	Texas Section	Text (40)
TX_LEAGUE	Texas League	Text (8)
TX_LABOR	Texas Labor	Text (8)
TX_TRACT_NUMBER	Texas Tract Number	Text (132)
TX_LOT_NUMBER	Texas Lot Number	Text (132)
MERIDIAN	Meridian	Numeric (Long)
TOWNSHIP	Township	Text (6)
TOWNSHIP_DIRECTION	Township Direction	Text (1)

Exported Name	Field Name	Type
RANGE	Range	Text (6)
RANGE_DIRECTION	Range Direction	Text (1)
SECTION	Section	Text (40)
SECTION_QUARTERS	Section Quarters	Text (132)
OFFSHORE_AREA	Offshore Area	Text (60)
OFFSHORE_BLOCK	Offshore Block	Text (60)
QUAD_NAME	Quad Name	Text (60)
REFERENCE_DETAIL	Reference Detail	Text (132)
REFERENCE_TYPE_CODE	Reference Type Code	Text (3)
REFERENCE_TYPE	Reference Type Description	Text (30)
objectid	Object ID	Object ID

3.1.4 File Geodatabase (FGDB) – Well Bottom Deleted, Spatial

This is the spatial layer related to well bottom hole locations that have been deleted from the IFS Tobin Well Product: *well_bottom_deleted*. The wells have been deleted because they are duplicates or temporary APIs or they are not part of the Oil & Gas process, i.e. cathodic wells, water wells – personal use, uranium wells, etc.

Exported Name	Field Name	Type
WellID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
API12	API Number (12)	Text (12)
STATE_CODE	API State Code	Numeric (Long)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Long)
COUNTY_NAME	County Name	Text (60)
RELATED_API	Related API	Text (30)
LEASE_NAME	Lease Name	Text (60)
WELL_NUM	Well Number	Text (60)
FIELD_NAME	Field Name	Text (30)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
BOREID	BOREID	Numeric (Long)

Exported Name	Field Name	Type
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
OBJECT_ID	Object ID	Object ID
SHAPE	Shape	Geometry

3.1.5 File Geodatabase (FGDB) – Well Path, Spatial

This is the spatial layer that establishes a primitive relationship between the well surface hole locations and the subsurface wellbore bottom hole locations for the IFS Tobin Well Product: *well_path*. This notional wellbore is only present for wells with a deviated, directional, or horizontal well profile.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
BOREID	BOREID	Numeric (Long)
STATE_CODE	API State Code	Numeric (Long)
COUNTY_CODE	API County Code	Numeric (Long)
API12	API Number (12)	Text(12)
OBJECT_ID	Object ID	Object ID
SHAPE	Shape	Geometry
SHAPE_length	Shape Length	Numeric (Double)

3.2 Shapefile (SHP)

The shapefile (SHP) deliverable for the IFS Tobin Well Product is composed of a spatial layer file and a tabular layer file related to the well surface hole locations, a spatial layer file and a tabular layer file related to the well bottom hole locations, a spatial layer file representing the path between the surface hole locations and bottom hole locations. There are also two (2) spatial layer files for deleted surface hole locations and bottom hole locations.

3.2.1 Shapefile (SHP) – Well Surface Location, Spatial

This is the spatial layer related to well surface hole locations for the IFS Tobin Well Product: *well_surface.shp*. This file includes pertinent “point in time” well header information including API 10, operator, ground elevation, etc., allowing for analysis of the spatial, physical, and lifecycle characteristics of the well.

Exported Name	Field Name	Type
FID	Feature ID	Object ID
Shape	SHAPE	Geometry
WELLID	WELLID	Numeric (Long)

Exported Name	Field Name	Type
API10	API Number (10)	Text (10)
WELLNUM	Well Number	Text (20)
LEASENAME	Lease Name	Text (60)
CURROPR	Current Operator	Text (60)
CURROPRAL	Aliased Current Operator	Text (60)
ORIGOPR	Original Operator	Text (60)
ORIGOPRAL	Aliased Original Operator	Text (60)
STATECD	API State Code	Numeric (Short)
STATENAME	State Name	Text (60)
COUNTYCD	API County Code	Numeric (Short)
COUNTYNAME	County Name	Text (60)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
SURFACEX	Surface X	Numeric (Double)
SURFACEY	Surface Y	Numeric (Double)
COORDSYSCD	Coordinate System Code	Numeric (Long)
ZONECD	Zone Code	Text (5)
CREATEDT	Created Date	Date (no timestamp)
UPDATEDT	Updated Date	Date (no timestamp)
PRMTDT	Permit Date	Date (no timestamp)
PRMTEXDT	Permit Expiration Date	Date (no timestamp)
SPUDDT	Spud Date	Date (no timestamp)
FRSTCMPLDT	First Completion Date	Date (no timestamp)
ACTSTATUS	Active Status Flag	Text (1)
PROPOSEDCD	Proposed Flag	Text (1)
GRNDEL	Ground Elevation	Numeric(Double)
GRNDELTYCD	Ground Elevation Type Code	Text (10)
GRNDELTYP	Ground Elevation Type Description	Text (40)
H2SFLAG	H2S Well	Text (1)
OBJECT_ID	Object ID	Numeric(Long)

3.2.1.1 Shapefile (SHP) – Well Surface Location, Tabular

This is the tabular layer related to well surface hole locations for the IFS Tobin Well Product: *well_surface_location.dbf*. This table includes survey and directional information. To determine location information for each well surface hole location, this file can be related or joined to the Well Surface Location spatial layer using the WELLID field.

Exported Name	Field Name	Type
OID	Object ID	Numeric (Long)
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
FIRSTDIR	First Direction	Text (15)
FRSTDIST	First Distance	Numeric (Long)
SCNDDIR	Second Direction	Text (15)
SCNDIST	Second Distance	Numeric (Long)
SPOTMDCODE	Spotting Method Code	Text (10)
SPOTMDDESC	Spotting Method Description	Text (30)
SPOTLOC	Spotted Location Description	Text (254)
SURVEYTYCD	Survey Type Code	Text (10)
SURVEYTY	Survey Type Description	Text (30)
MCD	Minor Civil Division	Text (60)
STDISTNM	State District Name	Text (60)
TXABSTRACT	Texas Abstract	Text (8)
TXGRANTEE	Texas Grantee	Text (60)
TXSURVEYNM	Texas Survey Name	Text (60)
TXBLOCK	Texas Block	Text (30)
TXSECTION	Texas Section	Text (40)
TXLEAGUE	Texas League	Text (8)
TXLABOR	Texas Labor	Text (8)
TXTRACTNUM	Texas Tract Number	Text (132)
TXLOTNUM	Texas Lot Number	Text (132)
MERIDIAN	Meridian	Numeric (Long)
TWNSHP	Township	Text (6)
TWNSHPDIR	Township Direction	Text (1)
RANGE	Range	Text (6)

Exported Name	Field Name	Type
RANGEDIR	Range Direction	Text (1)
SECTION	Section	Text (40)
SECTIONQTR	Section Quarters	Text (132)
OFFSHAREA	Offshore Area	Text (60)
OFFSHBLOCK	Offshore Block	Text (60)
QUADNAME	Quad Name	Text (60)
REFDETAIL	Reference Detail	Text (132)
REFTYPECD	Reference Type Code	Text (3)
REFTYPE	Reference Type Description	Text (30)
objectid	Object ID	Numeric (Long)

3.2.2 Shapefile (SHP) – Well Surface Deleted, Spatial

This is the spatial layer related to well surface hole locations that have been deleted from the IFS Tobin Well Product: *well_surface_deleted.shp*. The wells have been deleted because they are duplicates or temporary APIs or they are not part of the Oil & Gas process, i.e. cathodic wells, water wells – personal use, uranium wells, etc.

Exported Name	Field Name	Type
FID	Feature ID	Object ID
Shape	SHAPE	Geometry
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
STATECD	API State Code	Numeric (Long)
STATENAME	State Name	Text (60)
COUNTYCD	API County Code	Numeric (Long)
COUNTYNAME	County Name	Text (60)
RELATEAPI	Related API	Text (29)
LEASENAME	Lease Name	Text (60)
WELLNUM	Well Number	Text (60)
FIELDNAME	Field Name	Text (30)
CREATEDT	Created Date	Date (no timestamp)
UPDATEDT	Updated Date	Date (no timestamp)
LATITUDE	Latitude	Numeric (Double)

Exported Name	Field Name	Type
LONGITUDE	Longitude	Numeric (Double)
OBJECT_ID	Object ID	Numeric (Long)

3.2.3 Shapefile (SHP) – Well Bottom Location, Spatial

This is the spatial layer related to wellbore bottom hole locations for the IFS Tobin Well Product: *well_bottom.shp*. This file includes pertinent “point in time” well information including API 12, operator, permit, spud, and completion dates, KB and reference elevations, producing product, well status, etc., allowing for analysis of the spatial, physical, and lifecycle characteristics of the wellbore.

Exported Name	Field Name	Type
FID	Feature ID	Object ID
Shape	SHAPE	Geometry
WELLID	WELLID	Numeric (Long)
BOREID	BOREID	Numeric (Long)
API12	API Number (12)	Text (12)
API10	API Number (10)	Text (10)
SDTRKNUM	Sidetrack Number	Text (2)
REGLSNAME	Regulatory Lease Identifier	Text (60)
REGID	Regulatory ID Number	Text (30)
WELLNUM	Well Number	Text (20)
LEASENAME	Lease Name	Text (60)
CURROPER	Current Operator	Text (60)
CURROPERAL	Aliased Current Operator	Text (60)
ORIGOPER	Original Operator	Text (60)
ORIGOPERAL	Aliased Original Operator	Text (60)
STATECD	API State Code	Numeric (Long)
STATENAME	State Name	Text (60)
COUNTYCD	API County Code	Numeric (Long)
COUNTYNAME	County Name	Text (60)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
BH_X	Bottom Hole X	Numeric (Double)
BH_Y	Bottom Hole Y	Numeric (Double)

Exported Name	Field Name	Type
COORDSYSCD	Coordinate System Code	Numeric (Long)
ZONECD	Zone Code	Text (5)
CREATEDT	Created Date	Date (no timestamp)
UPDATEDT	Updated Date	Date (no timestamp)
PRMTDT	Permit Date	Date (no timestamp)
PRMTEXPDT	Permit Expiration Date	Date (no timestamp)
SPUDDT	Spud Date	Date (no timestamp)
FRSTCMPLDT	First Completion Date	Date (no timestamp)
CURRCMPLDT	Current Completion Date	Date (no timestamp)
FRSTPRODDT	First Production Date	Date (no timestamp)
LASTPRODDT	Last Production Date	Date (no timestamp)
PLUGDT	Plug Date	Date (no timestamp)
ACTSTATUS	Active Status Flag	Text (1)
PROPOSEDCD	Proposed Flag	Text (1)
FRSTEL	First Elevation	Numeric (Double)
FRSTELTYCD	First Elevation Type Code	Text (10)
FRSTELTY	First Elevation Type Description	Text (15)
SCNDEL	Second Elevation	Numeric (Double)
SCNDELTYCD	Second Elevation Type Code	Text (10)
SCNDELTY	Second Elevation Type Description	Text (15)
FIELDID	Field ID	Numeric (Long)
FIELDNAME	Field Name	Text (100)
REGFIELDID	Regulatory Field ID	Text (30)
REGFIELDNM	Regulatory Field Name	Text (50)
FORMNAME	Formation Name	Text (200)
PROPOSEDMD	Proposed MD	Numeric (Double)
KBELEV	KB Elevation	Numeric (Double)
TDMD	Total Depth	Numeric (Double)
TDTV	True Vertical Depth	Numeric (Double)
PLUGDEPTH	Plug Depth	Numeric (Double)
FNLSTATCD	Final Status Code	Text (3)

Exported Name	Field Name	Type
FNLSTATUS	Final Status Description	Text (30)
BORETYPECD	Well Bore Type Code	Text (3)
BORETYPE	Well Bore Type Description	Text (30)
BOREDIRCD	Well Direction Code	Text (3)
BOREDIR	Well Direction Description	Text (30)
PROFILECD	Well Profile Code	Text (3)
PROFILE	Well Profile Description	Text (30)
CLASSCD	Well Class Code	Text (3)
CLASS	Well Class Description	Text (30)
PRODUCTCD	Well Product Code	Text (3)
PRODUCT	Well Product Description	Text (30)
STATUSCODE	Well Status Code	Text (10)
STATUS	Well Status Description	Text (30)
OBJECT_ID	Object ID	Numeric (Long)

3.2.3.1 Shapefile (SHP) – Well Bottom Location, Tabular

This is the tabular layer related to wellbore bottom hole locations for the IFS Tobin Well Product: *well_bottom.dbf*. This table includes survey and directional information. To determine location information for each wellbore bottom hole location, this file can be related or joined to the Well Bottom Location spatial layer using the BOREID field.

Exported Name	Field Name	Type
OID	Object ID	OBJECT_ID
WELLID	WELLID	Integer
BOREID	BOREID	Integer
API10	API Number (10)	Text (10)
API12	API Number (12)	Text (12)
FIRSTDIR	First Direction	Text (15)
FRSTDIST	First Distance	Integer (round)
SCNDDIR	Second Direction	Text (15)
SCNDIST	Second Distance	Integer (round)
SPOTMDCODE	Spotting Method Code	Text (10)
SPOTMDDESC	Spotting Method Description	Text (30)

Exported Name	Field Name	Type
SPOTLOC	Spotted Location Description	Text (1024)
SURVEYTYCD	Survey Type Code	Text (10)
SURVEYTY	Survey Type Description	Text (30)
MCD	Minor Civil Division	Text (60)
STDISTNM	State District Name	Text (8)
TXABSTRACT	Texas Abstract	Text (60)
TXGRANTEE	Texas Grantee	Text (60)
TXSURVEYNM	Texas Survey Name	Text (30)
TXBLOCK	Texas Block	Text (40)
TXSECTION	Texas Section	Text (8)
TXLEAGUE	Texas League	Text (8)
TXLABOR	Texas Labor	Text (132)
TXTRACTNUM	Texas Tract Number	Text (132)
TXLOTNUM	Texas Lot Number	Integer
MERIDIAN	Meridian	Text (6)
TWNSHP	Township	Text (1)
TWNSHPDIR	Township Direction	Text (6)
RANGE	Range	Text (1)
RANGEDIR	Range Direction	Text (40)
SECTION	Section	Text (132)
SECTIONQTR	Section Quarters	Text (60)
OFFSHAREA	Offshore Area	Text (60)
OFFSHBLOCK	Offshore Block	Text (60)
QUADNAME	Quad Name	Text (132)
REFDETAIL	Reference Detail	Text (3)
REFTYPECD	Reference Type Code	Text (30)
REFTYPE	Reference Type Description	Numeric
objectid	Object ID	Numeric (Long)

3.2.4 Shapefile (SHP) – Well Bottom Deleted, Spatial

This is the spatial layer related to well bottom hole locations that have been deleted from the IFS Tobin Well Product: *well_bottom_deleted.shp*. The wells have been deleted because they are duplicates or temporary APIs or they are not part of the Oil & Gas process, i.e. cathodic wells, water wells – personal use, uranium wells, etc.

Exported Name	Field Name	Type
FID	Feature ID	OBJECT_ID
Shape	SHAPE	Geometry
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
API12	API Number (12)	Text (12)
STATECD	API State Code	Numeric (Long)
STATENAME	State Name	Text (60)
COUNTYCD	API County Code	Numeric (Long)
COUNTYNAME	County Name	Text (60)
RELATEAPI	Related API	Text (30)
LEASENAME	Lease Name	Text (60)
WELLNUM	Well Number	Text (60)
FIELDNAME	Field Name	Text (30)
CREATEDT	Created Date	Date (no timestamp)
UPDATEDT	Updated Date	Date (no timestamp)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
OBJECT_ID	Object ID	Numeric (Long)

3.2.5 Shapefile (SHP) – Well Path, Spatial

This is the spatial layer that establishes a primitive relationship between the well surface hole locations and the subsurface wellbore bottom hole locations for the IFS Tobin Well Product: *well_path.shp*. This notional wellbore is only present for wells with a deviated, directional, or horizontal well profile.

Exported Name	Field Name	Type
FID	Feature ID	OBJECT_ID
Shape	SHAPE	Geometry
WELLID	WELLID	Numeric (Long)

Exported Name	Field Name	Type
BOREID	BOREID	Numeric (Long)
STATECD	API State Code	Numeric (Long)
COUNTYCD	API County Code	Numeric (Long)
API12	API Number(12)	Text (12)
OBJECT_ID	Object ID	Numeric (Long)

3.3 Comma-Separated Values (CSV)

The comma-separated values (CSV) deliverable for the IFS Tobin Well Product is composed of two (2) tabular layer files related to the well surface hole locations and two (2) tabular layer files related to the well bottom hole locations. There are also two (2) tabular layer files for deleted surface hole locations and bottom hole locations.

3.3.1 Comma-Separated Values (CSV) – Well Surface, Tabular

This is the tabular layer related to well surface hole locations for the IFS Tobin Well Product: *well_surface.csv*. This file includes pertinent “point in time” well header information including API 10, operator, ground elevation, etc., allowing for analysis of the spatial, physical, and lifecycle characteristics of the well.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
WELL_NUM	Well Number	Text (20)
LEASE_NAME	Lease Name	Text (60)
CURRENT_OPERATOR	Current Operator	Text (60)
CURRENT_OPERATOR_ALIAS	Aliased Current Operator	Text (60)
ORIGINAL_OPERATOR	Original Operator	Text (60)
ORIGINAL_OPERATOR_ALIAS	Aliased Original Operator	Text (60)
STATE_CODE	API State Code	Numeric (Short)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Short)
COUNTY_NAME	County Name	Text (60)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
SURFACE_X	Surface X	Numeric (Double)
SURFACE_Y	Surface Y	Numeric (Double)

Exported Name	Field Name	Type
COORDINATE_SYSTEM_CODE	Coordinate System Code	Numeric (Long)
ZONE_CODE	Zone Code	Text (5)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
PERMIT_DATE	Permit Date	Date (no timestamp)
PERMIT_EXPIRY_DATE	Permit Expiration Date	Date (no timestamp)
SPUD_DATE	Spud Date	Date (no timestamp)
FIRST_COMPLETION_DATE	First Completion Date	Date (no timestamp)
ACTIVE_STATUS_FLAG	Active Status Flag	Text (1)
PROPOSED_FLAG	Proposed Flag	Text (1)
GROUND_ELEVATION	Ground Elevation	Numeric (Double)
GROUND_ELEVATION_TYPE_CODE	Ground Elevation Type Code	Text (10)
GROUND_ELEVATION_TYPE	Ground Elevation Type Description	Text (40)
H2S_FLAG	H2S Well	Text (1)

3.3.1.1 Comma-Separated Values (CSV) – Well Surface Location, Tabular

This is the tabular layer related to well surface hole locations for the IFS Tobin Well Product: *well_surface_location.csv*. This table includes survey and directional information. To determine location information for each well surface hole location, this file can be related or joined to the Well Surface Location spatial layer using the WELLID field.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
FIRST_DIRECTION	First Direction	Text (15)
FIRST_DISTANCE	First Distance	Numeric (Long)
SECOND_DIRECTION	Second Direction	Text (15)
SECOND_DISTANCE	Second Distance	Numeric (Long)
SPOTTED_METHOD_CODE	Spotting Method Code	Text (10)
SPOTTED_METHOD_DESCRIPTION	Spotting Method Description	Text (30)
SPOTTED_LOCATION	Spotted Location Description	Text (254)
SURVEY_TYPE_CODE	Survey Type Code	Text (10)
SURVEY_TYPE	Survey Type Description	Text (30)

Exported Name	Field Name	Type
MINOR_CIVIL_DIVISION	Minor Civil Division	Text (60)
STATE_DISTRICT_NAME	State District Name	Text (60)
TX_ABSTRACT	Texas Abstract	Text (8)
TX_GRANTEE	Texas Grantee	Text (60)
TX_SURVEY_NAME	Texas Survey Name	Text (60)
TX_BLOCK	Texas Block	Text (30)
TX_SECTION	Texas Section	Text (40)
TX_LEAGUE	Texas League	Text (8)
TX_LABOR	Texas Labor	Text (8)
TX_TRACT_NUMBER	Texas Tract Number	Text (132)
TX_LOT_NUMBER	Texas Lot Number	Text (132)
MERIDIAN	Meridian	Numeric (Long)
TOWNSHIP	Township	Text (6)
TOWNSHIP_DIRECTION	Township Direction	Text (1)
RANGE	Range	Text (6)
RANGE_DIRECTION	Range Direction	Text (1)
SECTION	Section	Text (40)
SECTION_QUARTERS	Section Quarters	Text (132)
OFFSHORE_AREA	Offshore Area	Text (60)
OFFSHORE_BLOCK	Offshore Block	Text (60)
QUAD_NAME	Quad Name	Text (60)
REFERENCE_DETAIL	Reference Detail	Text (132)
REFERENCE_TYPE_CODE	Reference Type Code	Text (3)
REFERENCE_TYPE	Reference Type Description	Text (30)

3.3.2 Comma-Separated Values (CSV) – Well Surface Deleted, Tabular

This is the tabular layer related to well surface hole locations that have been deleted from the IFS Tobin Well Product: *well_surface_deleted.csv*. The wells have been deleted because they are duplicates or temporary APIs or they are not part of the Oil & Gas process, i.e. cathodic wells, water wells – personal use, uranium wells, etc.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
STATE_CODE	API State Code	Numeric (Long)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Long)
COUNTY_NAME	County Name	Text (60)
RELATED_API	Related API	Text (29)
LEASE_NAME	Lease Name	Text (60)
WELL_NUM	Well Number	Text (60)
FIELD_NAME	Field Name	Text (30)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)

3.3.3 Comma-Separated Values (CSV) – Well Bottom, Tabular

This is the tabular layer related to wellbore bottom hole locations for the IFS Tobin Well Product: *well_bottom.csv*. This file includes pertinent “point in time” well information including API 12, operator, permit, spud, and completion dates, KB and reference elevations, producing product, well status, etc., allowing for analysis of the spatial, physical, and lifecycle characteristics of the wellbore.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
BOREID	BOREID	Numeric (Long)
API12	API Number (12)	Text (12)
API10	API Number (10)	Text (10)
SIDE_TRACK_NUMBER	Sidetrack Number	Text (2)
REGULATORY_LEASE_ID	Regulatory Lease Identifier	Text (60)
REGULATORY_ID	Regulatory ID Number	Text (30)

Exported Name	Field Name	Type
WELL_NUM	Well Number	Text (20)
LEASE_NAME	Lease Name	Text (60)
CURRENT_OPERATOR	Current Operator	Text (60)
CURRENT_OPERATOR_ALIAS	Aliased Current Operator	Text (60)
ORIGINAL_OPERATOR	Original Operator	Text (60)
ORIGINAL_OPERATOR_ALIAS	Aliased Original Operator	Text (60)
STATE_CODE	API State Code	Numeric (Long)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Long)
COUNTY_NAME	County Name	Text (60)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
BH_X	Bottom Hole X	Numeric (Double)
BH_Y	Bottom Hole Y	Numeric (Double)
COORDINATE_SYSTEM_CODE	Coordinate System Code	Numeric (Long)
ZONE_CODE	Zone Code	Text (5)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
PERMIT_DATE	Permit Date	Date (no timestamp)
PERMIT_EXPIRY_DATE	Permit Expiration Date	Date (no timestamp)
SPUD_DATE	Spud Date	Date (no timestamp)
FIRST_COMPLETION_DATE	First Completion Date	Date (no timestamp)
CURRENT_COMPLETION_DATE	Current Completion Date	Date (no timestamp)
FIRST_PRODUCTION_DATE	First Production Date	Date (no timestamp)
LAST_PRODUCTION_DATE	Last Production Date	Date (no timestamp)
PLUG_DATE	Plug Date	Date (no timestamp)
ACTIVE_STATUS_FLAG	Active Status Flag	Text (1)
PROPOSED_FLAG	Proposed Flag	Text (1)
FIRST_ELEVATION	First Elevation	Numeric (Double)
FIRST_ELEVATION_TYPE_CODE	First Elevation Type Code	Text (10)
FIRST_ELEVATION_TYPE	First Elevation Type Description	Text (15)

Exported Name	Field Name	Type
SECOND_ELEVATION	Second Elevation	Numeric (Double)
SECOND_ELEVATION_TYPE_CODE	Second Elevation Type Code	Text (10)
SECOND_ELEVATION_TYPE	Second Elevation Type Description	Text (15)
FIELD_ID	Field ID	Numeric (Long)
FIELD_NAME	Field Name	Text (100)
REGULATORY_FIELD_ID	Regulatory Field ID	Text (30)
REGULATORY_FIELD_NAME	Regulatory Field Name	Text (50)
FORMATION_NAME	Formation Name	Text (200)
PROPOSED_MD	Proposed MD	Numeric (Double)
KB_ELEVATION	KB Elevation	Numeric (Double)
TOTAL_DEPTH_MD	Total Depth	Numeric (Double)
TOTAL_DEPTH_TVD	True Vertical Depth	Numeric (Double)
PLUG_DEPTH	Plug Depth	Numeric (Double)
FINAL_STATUS_CODE	Final Status Code	Text (3)
FINAL_STATUS	Final Status Description	Text (30)
BORE_TYPE_CODE	Well Bore Type Code	Text (3)
BORE_TYPE	Well Bore Type Description	Text (30)
BORE_DIRECTION_CODE	Well Direction Code	Text (3)
BORE_DIRECTION	Well Direction Description	Text (30)
PROFILE_CODE	Well Profile Code	Text (3)
PROFILE	Well Profile Description	Text (30)
CLASS_CODE	Well Class Code	Text (3)
CLASS	Well Class Description	Text (30)
PRODUCT_CODE	Well Product Code	Text (3)
PRODUCT	Well Product Description	Text (30)
STATUS_CODE	Well Status Code	Text (10)
STATUS	Well Status Description	Text (30)

3.3.3.1 Comma-Separated Values (CSV) – Well Bottom Location, Tabular

This is the tabular layer related to wellbore bottom hole locations for the IFS Tobin Well Product: *well_bottom_location.csv*. This table includes survey and directional information. To determine location information for each wellbore bottom hole location, this file can be related or joined to the Well Bottom Location spatial layer using the BOREID field.

Exported Name	Field Name	Type
WELLID	WELLID	Integer
BOREID	BOREID	Integer
API10	API Number (10)	Text (10)
API12	API Number (12)	Text (12)
FIRST_DIRECTION	First Direction	Text (15)
FIRST_DISTANCE	First Distance	Integer (round)
SECOND_DIRECTION	Second Direction	Text (15)
SECOND_DISTANCE	Second Distance	Integer (round)
SPOTTED_METHOD_CODE	Spotting Method Code	Text (10)
SPOTTED_METHOD_DESCRIPTION	Spotting Method Description	Text (30)
SPOTTED_LOCATION	Spotted Location Description	Text (1024)
SURVEY_TYPE_CODE	Survey Type Code	Text (10)
SURVEY_TYPE	Survey Type Description	Text (30)
MINOR_CIVIL_DIVISION	Minor Civil Division	Text (60)
STATE_DISTRICT_NAME	State District Name	Text (8)
TX_ABSTRACT	Texas Abstract	Text (60)
TX GRANTEE	Texas Grantee	Text (60)
TX_SURVEY_NAME	Texas Survey Name	Text (30)
TX_BLOCK	Texas Block	Text (40)
TX_SECTION	Texas Section	Text (8)
TX_LEAGUE	Texas League	Text (8)
TX_LABOR	Texas Labor	Text (132)
TX_TRACT_NUMBER	Texas Tract Number	Text (132)
TX_LOT_NUMBER	Texas Lot Number	Integer
MERIDIAN	Meridian	Text (6)
TOWNSHIP	Township	Text (1)
TOWNSHIP_DIRECTION	Township Direction	Text (6)

Exported Name	Field Name	Type
RANGE	Range	Text (1)
RANGE_DIRECTION	Range Direction	Text (40)
SECTION	Section	Text (132)
SECTION_QUARTERS	Section Quarters	Text (60)
OFFSHORE_AREA	Offshore Area	Text (60)
OFFSHORE_BLOCK	Offshore Block	Text (60)
QUAD_NAME	Quad Name	Text (132)
REFERENCE_DETAIL	Reference Detail	Text (3)
REFERENCE_TYPE_CODE	Reference Type Code	Text (30)
REFERENCE_TYPE	Reference Type Description	Numeric

3.3.4 Comma-Separated Values (CSV) – Well Bottom Deleted, Tabular

This is the tabular layer related to well bottom hole locations that have been deleted from the IFS Tobin Well Product: *well_bottom_deleted.csv*. The wells have been deleted because they are duplicates or temporary APIs or they are not part of the Oil & Gas process, i.e. cathodic wells, water wells – personal use, uranium wells, etc.

Exported Name	Field Name	Type
WELLID	WELLID	Numeric (Long)
API10	API Number (10)	Text (10)
API12	API Number (12)	Text (12)
STATE_CODE	API State Code	Numeric (Long)
STATE_NAME	State Name	Text (60)
COUNTY_CODE	API County Code	Numeric (Long)
COUNTY_NAME	County Name	Text (60)
RELATED_API	Related API	Text (30)
LEASE_NAME	Lease Name	Text (60)
WELL_NUM	Well Number	Text (60)
FIELD_NAME	Field Name	Text (30)
CREATED_DATE	Created Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)

3.4 Comma-Separated Values Combined (CSV)

The comma-separated values combined (CSV) deliverable for the IFS Tobin Well Product is composed of a single tabular layer file. This file contains a single record for each well with both limited surface and bottom hole information. This is the single tabular layer for the IFS Tobin Well Product: *well_combined.csv*.

Exported Name	Field Name	Type
API12	API Number (12)	Text (12)
CURRENT_OPERATOR_ALIAS	Aliased Current Operator	Text (60)
ORIGINAL_OPERATOR_ALIAS	Aliased Original Operator	Text (60)
LEASE_NAME	Lease Name	Text (60)
WELL_NUM	Well Number	Text (20)
KB_ELEVATION	KB Elevation	Numeric (Double)
TOTAL_DEPTH_MD	Total Depth	Numeric (Double)
TOTAL_DEPTH_TVD	True Vertical Depth	Numeric (Double)
LATITUDE	Latitude	Numeric (Double)
LONGITUDE	Longitude	Numeric (Double)
SURFACE_X	Surface X	Numeric (Double)
SURFACE_Y	Surface Y	Numeric (Double)
BH_LATITUDE	Latitude	Numeric (Double)
BH_LONGITUDE	Longitude	Numeric (Double)
BH_X	Bottom Hole X	Numeric (Double)
BH_Y	Bottom Hole Y	Numeric (Double)
COORDINATE_SYSTEM_CODE	Coordinate System Code	Numeric (Long)
PRODUCT_CODE	Well Product Code	Text (3)
STATUS_CODE	Well Status Code	Text (10)
FINAL_STATUS_CODE	Final Status Code	Text (3)
FIRST_COMPLETION_DATE	First Completion Date	Date (no timestamp)
FIELD_NAME	Field Name	Text (100)
SPUD_DATE	Spud Date	Date (no timestamp)
ORIGINAL_OPERATOR	Original Operator	Text (60)
CURRENT_OPERATOR	Current Operator	Text (60)
SECTION	Section	Text (40)
TOWNSHIP	Township	Text (6)
TOWNSHIP_DIRECTION	Township Direction	Text (1)

Exported Name	Field Name	Type
RANGE	Range	Text (6)
RANGE_DIRECTION	Range Direction	Text (1)
MERIDIAN	Meridian	Numeric (Long)
MINOR_CIVIL_DIVISION	Minor Civil Division	Text (60)
STATE_DISTRICT_NAME	State District Name	Text (60)
SURVEY_TYPE	Survey Type Description	Text (30)
TX_ABSTRACT	Texas Abstract	Text (8)
TX_LOT_NUMBER	Texas Lot Number	Text (132)
TX_TRACT_NUMBER	Texas Tract Number	Text (132)
TX_BLOCK	Texas Block	Text (30)
FIRST_DISTANCE	First Distance	Numeric (Long)
FIRST_DIRECTION	First Direction	Text (15)
SECOND_DISTANCE	Second Distance	Numeric (Long)
SECOND_DIRECTION	Second Direction	Text (15)
BH_SECTION	Section	Text (40)
BH_TOWNSHIP	Township	Text (6)
BH_TOWNSHIP_DIRECTION	Township Direction	Text (1)
BH_RANGE	Range	Text (6)
BH_RANGE_DIRECTION	Range Direction	Text (1)
BH_MERIDIAN	Meridian	Numeric (Long)
BH_MINOR_CIVIL_DIVISION	Minor Civil Division	Text (60)
BH_STATE_DISTRICT_NAME	State District Name	Text (60)
BH_SURVEY_TYPE	Survey Type Description	Text (30)
BH_TX_ABSTRACT	Texas Abstract	Text (8)
BH_TX_LOT_NUMBER	Texas Lot Number	Text (132)
BH_TX_TRACT_NUMBER	Texas Tract Number	Text (132)
BH_TX_BLOCK	Texas Block	Text (30)
PROPOSED_MD	Proposed MD	Numeric (Double)
WCS_UPDATE_DATE	WCS Update Date	Date (no timestamp)
UPDATE_DATE	Updated Date	Date (no timestamp)
CREATED_DATE	Created Date	Date (no timestamp)

Exported Name	Field Name	Type
H2S_FLAG	H2S Well	Text (1)
GROUND_ELEVATION	Ground Elevation	Numeric (Double)
GROUND_ELEVATION_TYPE	Ground Elevation Type Description	Text (40)
GROUND_ELEVATION_SOURCE	Ground Elevation Source	Text (12)
KB_ELEVATION_SOURCE	KB Elevation Source	Text (12)
COMPLETED_FORMATION	Formation Name	Text (200)

3.5 Well Coordinate Service (WCS)

Well Coordinate Service (WCS) is a legacy ASC-II file format available to users for delivery of the IFS Tobin Well Product.

For more information about IFS Tobin's Well Coordinate Service (WCS) file format, download the "WCS Format Specification" document available on the Tobin Data Portal or contact IFS Tobin Support at 844-732-2472, option 6, or enr.tobinsupport@ifs.com.

4. Code Value Tables and Additional Reference Information

4.1 Elevation Type Code Values

Note: This table contains code values for First, Second, and Ground Elevations.

Code	Description	Code	Description
CB	Center Line Braden Head Outlet	MT	Drill Floor Mat
CF	Casing Head Flange	PT	Pipe Top
DF	Derrick Floor	RB	Rotary Bushing
EG	Estimated Ground Elevation	RF	Rig Floor
ES	Estimated	RT	Rotary Table
GR	Ground Elevation	TS	Topo Sheet
KB	Kelly Bushing	UNK	Unknown

4.2 Final Status Code Values

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
ABD	Abandoned Well	Abandoned				Multiple
ABD	Abandoned Well	Abandoned				
ABD	Abandoned Well	Abandoned			Observation	
ABD	Abandoned Well	Abandoned			Pilot Hole	
ABD	Abandoned Well	Abandoned			Service	
ABD	Abandoned Well	Abandoned			Strat Test	
ABD	Abandoned Well	Abandoned			Inject Assist	
ABD	Abandoned Well	Abandoned			Disposal	
ABD	Abandoned Well	Orphan				
AGW	Abandoned Gas	Dry and Abandoned	Producer	Gas-Water		
AGW	Abandoned Gas	Dry and Abandoned	Producer	Gas		
AGW	Abandoned Gas	Dry and Abandoned	Producer	Coalbed Methane		
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Gas-Water		
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Gas		
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Coalbed Methane		
AGW	Abandoned Gas	Abandoned	Producer	Methane		
AGW	Abandoned Gas	Abandoned	Producer	Gas-Water		
AGW	Abandoned Gas	Abandoned	Producer	Gas		
AGW	Abandoned Gas	Abandoned	Producer	Coalbed Methane		
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Condensate		
AGW	Abandoned Gas	Abandoned	Producer	Condensate		
AGW	Abandoned Gas	Dry and Abandoned	Producer	Condensate		
AGW	Abandoned Gas	Dry and Abandoned	Producer	Methane		
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Coalbed Methane		Multiple
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Condensate		Multiple

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Gas		Multiple
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Gas-Water		Multiple
AGW	Abandoned Gas	Plugged and Abandoned	Producer	Methane		Multiple
AOW	Abandoned Oil	Abandoned	Producer	Oil-Water		
AOW	Abandoned Oil	Abandoned	Producer	Oil-Gas		
AOW	Abandoned Oil	Dry and Abandoned	Producer	Oil-Gas		
AOW	Abandoned Oil	Abandoned	Producer	Oil		
AOW	Abandoned Oil	Plugged and Abandoned	Producer	Oil-Water		
AOW	Abandoned Oil	Dry and Abandoned	Producer	Oil-Water		
AOW	Abandoned Oil	Plugged and Abandoned	Producer	Oil-Gas		
AOW	Abandoned Oil	Plugged and Abandoned	Producer	Oil		
AOW	Abandoned Oil	Plugged and Abandoned	Producer	Oil-Gas		Multiple
AOW	Abandoned Oil	Plugged and Abandoned	Producer	Oil-Water		Multiple
AOW	Abandoned Oil	Plugged and Abandoned	Producer	Oil		Multiple
AOW	Abandoned Oil	Dry and Abandoned	Producer	Oil		
CBM	Coalbed Methane	Active	Producer	Coalbed Methane		
CNC	Cancel	Abandoned Location			Observation	
CNC	Cancel	Abandoned Location			Pilot Hole	
CNC	Cancel	Abandoned Location			Service	
CNC	Cancel	Abandoned Location			Strat Test	
CNC	Cancel	Cancelled			Inject Assist	
CNC	Cancel	Cancelled				Multiple
CNC	Cancel	Abandoned Location				Multiple

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
CNC	Cancel	Abandoned Location				
CNC	Cancel	Abandoned Location			Inject Assist	
CNC	Cancel	Abandoned Location			Disposal	
CNC	Cancel	Cancelled				
CNC	Cancel	Cancelled			Disposal	
CNC	Cancel	Cancelled			Observation	
CNC	Cancel	Cancelled			Pilot Hole	
CNC	Cancel	Cancelled			Service	
CNC	Cancel	Cancelled			Strat Test	
CO2	Carbon Dioxide	Active	Producer	Carbon Dioxide		
D&A	Dry and Abandoned	Dry and Abandoned	Producer	Dry		
D&A	Dry and Abandoned	Dry and Abandoned			Strat Test	
D&A	Dry and Abandoned	Dry and Abandoned				
D&A	Dry and Abandoned	Dry and Abandoned	Producer	Unknown		
D&A	Dry and Abandoned	Completed	Producer	Dry		
D&A	Dry and Abandoned	Completed	Unknown	Dry		
DAG	DA Gas Show	Dry and Abandoned		Dry Gas Show		
DAO	DA Oil Show	Dry and Abandoned		Dry Oil Show		
DAX	DA Oil/Gas Show	Dry and Abandoned		Dry Oil/Gas Show		
DIS	Disposal Well				Disposal	
DRL	Drilling	Drilling				
DUC	Drilling, Uncompleted	Drilling, Uncompleted				
EXP	Expired	Expired			Inject Assist	
EXP	Expired	Expired				Multiple
EXP	Expired	Expired				
EXP	Expired	Expired			Strat Test	

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
EXP	Expired	Expired			Observation	
EXP	Expired	Expired			Pilot Hole	
EXP	Expired	Expired			Service	
EXP	Expired	Expired			Disposal	
GAS	Gas	Active	Producer	Gas-Water		
GAS	Gas	Active	Producer	Gas		
GAS	Gas	Active	Producer	Condensate		
GSW	Gas Storage	Active	Huff and Puff	Gas-Water		
GSW	Gas Storage	Active	Huff and Puff	Coalbed Methane		
GSW	Gas Storage	Active	Huff and Puff	Methane		
GSW	Gas Storage	Active	Huff and Puff	Condensate		
GSW	Gas Storage	Active	Huff and Puff	Gas		
HEL	Helium	Active	Producer	Helium		
INJ	Injection Well	Active	Injection		Inject Assist	
INJ	Injection Well	Active	Injection			
INS	In Situ Combustion	Active	Huff and Puff	Air	Inject Assist	
IOD	Iodine	Active	Producer	Iodine		
J&A	Junk	Junked and Abandoned			Disposal	
J&A	Junk	Junked and Abandoned			Observation	
J&A	Junk	Junked and Abandoned			Pilot Hole	
J&A	Junk	Junked and Abandoned				
J&A	Junk	Junked and Abandoned			Strat Test	
J&A	Junk	Junked and Abandoned			Inject Assist	
J&A	Junk	Junked and Abandoned				Multiple
J&A	Junk	Junked and Abandoned			Service	
LOC	Location Permit	Completed			Inject Assist	

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
LOC	Location Permit	Permitted Location				Multiple
LOC	Location Permit	Completed			Pilot Hole	
LOC	Location Permit	Completed			Service	
LOC	Location Permit	Working Over			Disposal	
LOC	Location Permit	Working Over			Inject Assist	
LOC	Location Permit	Working Over			Observation	
LOC	Location Permit	Working Over			Pilot Hole	
LOC	Location Permit	Working Over			Service	
LOC	Location Permit	Working Over			Strat Test	
LOC	Location Permit	Permitted Location			Inject Assist	
LOC	Location Permit	Completed			Disposal	
LOC	Location Permit	Completed				Multiple
LOC	Location Permit	Working Over				Multiple
LOC	Location Permit	Working Over				
LOC	Location Permit	Completed				
LOC	Location Permit	Permitted Location				
LOC	Location Permit	Permitted Location			Disposal	
LOC	Location Permit	Permitted Location			Observation	
LOC	Location Permit	Permitted Location			Pilot Hole	
LOC	Location Permit	Permitted Location			Service	
LOC	Location Permit	Permitted Location			Strat Test	
LOC	Location Permit	Completed			Observation	
M-G	Multi-Gas	Active	Producer	Gas-Water		Multiple
M-G	Multi-Gas	Active	Producer	Condensate		Multiple
M-G	Multi-Gas	Active	Producer	Gas		Multiple

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
M-G	Multi-Gas	Active	Producer	Methane		Multiple
M-G	Multi-Gas	Active	Producer	Coalbed Methane		Multiple
M-O	Multi-Oil	Active	Producer	Oil		Multiple
M-O	Multi-Oil	Active	Producer	Oil-Water		Multiple
MET	Methane	Active	Producer	Methane		
MOG	Oil & Gas	Active	Producer	Oil-Gas		
N/A	Not Available		Producer	Air		
N/A	Not Available					
N/A	Not Available	Dry and Abandoned			Disposal	
N/A	Not Available	Dry and Abandoned			Observation	
N/A	Not Available	Dry and Abandoned			Service	
N/A	Not Available	Junked				
N/A	Not Available	Junked			Disposal	
N/A	Not Available	Junked			Inject Assist	
N/A	Not Available	Junked			Observation	
N/A	Not Available	Junked			Pilot Hole	
N/A	Not Available	Junked			Service	
N/A	Not Available	Junked			Strat Test	
N/A	Not Available	Plugged			Disposal	
N/A	Not Available	Plugged			Inject Assist	
N/A	Not Available	Plugged			Observation	
N/A	Not Available	Plugged			Pilot Hole	
N/A	Not Available	Plugged			Service	
N/A	Not Available	Plugged			Strat Test	
N/A	Not Available	Plugged				
N/A	Not Available	Dry and Abandoned			Inject Assist	
N/A	Not Available	Dry				Multiple
N/A	Not Available	Dry and Abandoned				Multiple
N/A	Not Available	Dry and Abandoned	Injection			
N/A	Not Available	Dry			Disposal	

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
N/A	Not Available	Dry			Observation	
N/A	Not Available	Dry			Pilot Hole	
N/A	Not Available	Dry			Service	
N/A	Not Available	Dry			Strat Test	
N/A	Not Available	Dry	Injection			
N/A	Not Available	Active		Dry		
N/A	Not Available	Dry	Producer			
N/A	Not Available	Unknown				
N/A	Not Available	Unknown				Multiple
N/A	Not Available	Junked				Multiple
N/A	Not Available	Plugged				Multiple
N/A	Not Available	Dry			Inject Assist	
N/A	Not Available	Dry and Abandoned			Pilot Hole	
N/A	Not Available	Active	Producer	Dry		
OBS	Observation				Observation	
OIL	Oil	Active	Huff and Puff	Oil-Gas	Inject Assist	
OIL	Oil	Active	Producer	Oil		
OIL	Oil	Active	Producer	Oil-Water		
OIL	Oil	Active	Huff and Puff	Oil	Inject Assist	
OSW	Oil Storage	Active	Huff and Puff	Oil	Storage	
OSW	Oil Storage	Active	Huff and Puff	Oil-Water	Storage	
OSW	Oil Storage	Active	Huff and Puff	Oil-Gas	Storage	
P&A	Plug Abandoned	Plugged and Abandoned				
P&A	Plug Abandoned	Plugged and Abandoned			Inject Assist	
P&A	Plug Abandoned	Plugged and Abandoned			Disposal	
P&A	Plug Abandoned	Plugged and Abandoned			Observation	
P&A	Plug Abandoned	Plugged and Abandoned			Pilot Hole	

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
P&A	Plug Abandoned	Plugged and Abandoned			Service	
P&A	Plug Abandoned	Plugged and Abandoned				
P&A	Plug Abandoned	Plugged and Abandoned	Huff and Puff		Geothermal	
P&A	Plug Abandoned	Plugged and Abandoned			Strat Test	
PLT	Pilot				Pilot Hole	
POT	Potash	Active	Producer	Potash		
PSE	Pseudo Status	Active		Unknown		
SER	Service Well				Service	
SI	Shut In	Shut In			Inject Assist	
SI	Shut In	Shut In			Observation	
SI	Shut In	Shut In			Disposal	
SI	Shut In	Shut In			Service	
SI	Shut In	Shut In			Pilot Hole	
SI	Shut In	Shut In			Strat Test	
SI	Shut In	Shut In				
SI	Shut In	Shut In				Multiple
SLT	Salt	Active	Producer	Salt		
STE	Geo. Steam	Active	Producer	Geo Steam		
STE	Geo. Steam		Huff and Puff		Geothermal	
STE	Geo. Steam	Active	Huff and Puff	Geo Steam	Geothermal	
STR	Strat Test Well				Strat Test	
STR	Strat Test Well				Core Test	
SUL	Sulphur	Active	Producer	Sulphur		
SUS	Suspended	Suspended			Strat Test	
SUS	Suspended	Suspended			Service	
SUS	Suspended	Suspended			Pilot Hole	
SUS	Suspended	Suspended			Observation	
SUS	Suspended	Suspended			Disposal	
SUS	Suspended	Suspended				
SUS	Suspended	Suspended			Inject Assist	
SUS	Suspended	Suspended				Multiple

Final Status Code	Description	Well Status	Direction	Product	Class	Completion Count
SWD	Salt Water Disposal	Active	Injection	Water-Brine		
T&A	Temp Abandoned	Temporary Abandoned			Inject Assist	
T&A	Temp Abandoned	Temporary Abandoned				Multiple
T&A	Temp Abandoned	Temporary Abandoned			Disposal	
T&A	Temp Abandoned	Temporary Abandoned				
T&A	Temp Abandoned	Temporary Abandoned			Pilot Hole	
T&A	Temp Abandoned	Temporary Abandoned			Service	
T&A	Temp Abandoned	Temporary Abandoned			Strat Test	
T&A	Temp Abandoned	Temporary Abandoned			Observation	
WTR	Water Well	Active	Producer	Water-Brine		
WTR	Water Well	Active	Producer	Water		

4.3 Spotted Method Code Values

Code	Description
AL	Along A Line
BD	Bearing Distance
C	Center
DD	Direction Distance
KP	Known Point
P	Perpendicular
U	Unknown

4.4 Reference Type Code Values

Code	Description
A	API
L	LEASE
M	MONUMENT
O	OWN
R	RUR SUB
RP	REFERENCE POINT
S	SURVEY
SL	SURFACE
U	UNKNOWN
W	WELL NUMBER

4.5 Survey Type Code Values

Code	Description
J	Jeffersonian
C	Carter
T	Texas
S	State Waters
F	Federal Waters
E	Eastern States

4.6 Well Bore Type Code Values

Code	Description
OH	Original Hole
ST	Sidetrack
RD	Re-drill
BP	Bypass

4.7 Well Class Code Values

Code	Description	Code	Description
CT	Core Test	P	Pilot Hole
D	Development	S	Service
DP	Deeper Pool	SP	Shallower Pool
DSP	Disposal	ST	Strat Test
ET	Extension Test	STO	Storage
G	Geothermal	W	Wildcat
I	Inject Assist	WP	Wildcat Pool
O	Observation		

4.8 Well Direction Code Values

Code	Description
I	Injection
H	Huff and Puff
P	Producer
U	Unknown

4.9 Well Product Code Values

Code	Description	Code	Description
A	Air	I	Iodine
CD	Condensate	M	Methane
CM	Coalbed Methane	O	Oil
CO2	Carbon Dioxide	OG	Oil-Gas
D	Dry	OW	Oil-Water
DG	Dry Gas Show	P	Potash
DO	Dry Oil Show	S	Salt
DX	Dry Oil/Gas Show	SU	Sulphur
G	Gas	SW	Water-Brine
GEO	Geo Steam	U	Unknown
GW	Gas-Water	W	Water
H	Helium		

4.10 Well Profile Code Values

Code	Description
V	Vertical
D	Directional
H	Horizontal
U	Unknown

4.11 Well Status Code Values

Code	Description	Code	Description
AB	Abandoned	JA	Junked and Abandoned
ACT	Active	OR	Orphan
AL	Abandoned Location	P	Plugged
C	Completed	PA	Plugged and Abandoned
CA	Cancelled	PL	Permitted Location
D	Dry	S	Shut In
DA	Dry and Abandoned	SUS	Suspended
DR	Drilling	TA	Temporarily Abandoned
DUC	Drilled, Uncompleted	U	Unknown
E	Expired	WO	Working Over
J	Junked		

4.12 Data Source Code Values

Code	Description	Code	Description
CLIENT	Client	ST_KYKGS	KY Geological Survey
DONOTUSE	Place Holder Value	ST_LADNR	LA Dept of Natural Resources
DWS	Directional Well Survey	ST_MIDEQ	Michigan Dept of Environmental Quality
FD_BLM	US Bureau of Land Management	ST_MO	Missouri
FD_BOEMRE	Fed Offshore	ST_MSOGB	Mississippi Oil and Gas Board
FD_USGS	US Geological Survey	ST_MTBOGC	MT Board of Oil & Gas
HIST	Historical Ticket	ST_NDOGD	North Dakota Oil and Gas Division

Code	Description	Code	Description
HPDI	HPDI	ST_NEOGCC	NE Oil & Gas Conservation Commission
NWH	NWH	ST_NMOCD	NM Oil Conservation Division
P2_CALCFGD	IFS Tobin Calculated from Ground Elevation	ST_NVDM	Nevada Division of Mines and Minerals
P2ES	IFS Tobin	ST_NYDEC	New York DEC
SCOUT	Scout Ticket	ST_OHDNR	OH Dept of Natural Resources
ST_AKOGCC	AK Oil & Gas Conservation Commission	ST_OKOCC	Oklahoma Commission
ST_ALSOGB	AL State Oil & Gas Board	ST_OR	Oregon
ST_ARAOGC	AR Oil & Gas Commission	ST_PA	Pennsylvania
ST_AZOGCC	Arizona Oil & Conservation Commission	ST_PADCNR	Pennsylvania Dept of Cons and Nat Res
ST_CADOGGR	California department of Oil, Gas	ST_PADEP	PA Dept of Environmental Protection
ST_COOGC	CO Oil & Gas Commission	ST_SDDENR	South Dakota Dept of Env and Nat Res
ST_FDOGC	Federal Oil and Gas Commission	ST_TN	Tennessee
ST_FL	Florida	ST_TXGLO	TX General Land Office
ST_IAGS	Iowa Geological Survey	ST_TXRRC	TX Railroad Commission
ST_ID	Idaho	ST_UTOCD	UT Div of Oil, Gas & Mining
ST_IL	Illinois	ST_VADMME	VA Dept of Mines, Minerals and Energy
ST_ILDNR	Illinois Dept of Natural Resources	ST_WA	Washington
ST_INDNR	Indiana Dept of Natural Resources	ST_WI	Wisconsin
ST_INIGS	Indiana Geological Survey	ST_WVGS	WV Geological and Economic Survey

Code	Description	Code	Description
ST_ISGS	Illinois Oil and Gas Resources	ST_WYOGCC	WY Oil & Gas Conservation Commission
ST_KSKGS	KS Geological Survey	STATE	State
ST_KYDNR	Kentucky Dept of Natural Resources	UNKNOWN	Unknown

4.13 North American Datum (NAD) State Plane Coordinate System (SPCS) Zone Codes

Information about SPCS Zone Codes is available on the ESRI ArcGIS Hub website:

[USA State Plane Zones NAD27](#)

[USA State Plane Zones NAD83](#)

4.14 Additional Reference Information

For more information about the Public Land Survey System (PLSS) legal descriptions, visit the following link on the Bureau of Land Management's website:

<https://www.blm.gov/sites/blm.gov/files/SpecificationsForDescriptionsOfLand.pdf>

For more Information about the Original Texas Land Survey System, request the "Terms Pertaining to the Tobin Original Texas Land Survey Data" document by contacting IFS Tobin Support at 844-732-2472, option 6, or enr.tobinsupport@ifs.com.